



CSCI 6371 Robotics Programming Syllabus

Course Information

MON 06:30 PM 09:00 PM Edinburg EENGR 1.272

Traditional Face-to-Face

Instructor Information

Qi Lu

Email: qi.lu@utrgv.edu

Office Location

EIEAB 3.216

Office Hours: Mon./Wed. 12:30 - 2:00pm

Welcome to my class. My name is Luki. It is the reverse of my first name and last name LuQi. It is hard to pronounce Qi in English. So, I made my English name Luki. It is much easier. The teacher's mindset influences students. An inclusive teaching environment is very important. I understand that a "growth mindset" can significantly improve students' performance. Everyone will make mistakes and can improve if you believe that you can improve by learning. Instead of saying "If you struggle in my class, remember that not everybody is a "CS person" who can easily learn and apply the right facts and techniques for problems", I would like to say "Everyone makes mistakes when they're learning new and more challenging topics. In my class, we'll celebrate those mistakes. Your mistakes tell me that you are making progress and that I need to have a better explanation of this topic".

Discord account: luki2485; We can discuss questions on Discord.

Teaching Assistant: Sara Mahjabin Hridita

Email: sara.hridita01@utrgv.edu

Location: EIEAB 2.204 MARS research lab.

Office hours: Tue/Wed, 11 – 12:30 pm. Will also be held virtually on Discord.

Discord account: hena_39250

Grading Schema

Activity	Percentage
Attendance	5%
Homework Assignments	35%
Midterm Exams	40%
Final project	20%
Total Values	100%

Course Description

This class covers the fundamentals of autonomous mobile robots and basic programming for mobile robots. The topics include locomotion, sensors, image processing, localization, motion planning, and swarm robotics. It also provides several practical sessions with hands-on algorithm development in coordinating multi-robot systems and robot swarms. You will have the chance to export your algorithms into physical robots if your algorithms perform efficiently in a simulation. The students will undertake assignments in theory, mini projects in programming, and presentations in their final projects.

Credit Hours: 3

Core Curriculum Component Area:

N/A

Optional - Special Designations

N/A

Tentative Calendar of Activities

Tentative course schedule		
Weeks	Topics	Assignments
Week 1: (Aug. 26 – 28)	No class	

Tentative course schedule

Week 2: (Aug. 31 – Sep. 4)	Introduction and motivation Lecture overview Locomotion & Perception	Assignment 1
Week 3: (Sep. 7 – 11)	Computer vision	
Week 4: (Sep. 14 – 18)	Image processing	Assignment 2; Project 1
Week 5: (Sep. 21 – 25)	Image processing	Assignment 3
Week 6: (Sep. 28 – Oct. 2)	Image processing	Midterm 1 review
Week 7: (Oct. 5 – 9)	Place recognition & Line Extraction	Midterm 1
Week 8: (Oct. 12 – 16)	Localization & SLAM Motion planning	Assignment 4
Week 9: (Oct. 19 – 23)	Swarm robotics	Midterm 2 review Assignment 5
Week 10: (Oct. 26 – 30)	Multi-robot system I	Final project discussion Midterm 2
Week 11: (Nov. 2 – 6)	Multi-robot system II	
Week 12: (Nov. 9 – 13)	Multi-robot system II	
Week 13: (Nov. 16 – 20)	Robot learning	
Week 14: (Nov. 23 – 27) Thanksgiving	Final exam review, Q&A	
Week 15: (Nov. 30 – Dec. 4)	Project presentation	
Week 16: (Dec. 7 – 11)	Project presentation	

Program Learning Objectives

N/A

Learning Objective for Core Curriculum Requirements

By the end of the course, students should

- Develop programming skills in robotics-related topics.
- Demonstrate proficiency with robot simulation and design.
- Understand how to control a mobile robot.
- Understand how a robot plans action.
- Know coordination paradigms in systems of multiple robots.
- Know classical multi-robot problems and their solution methods.

Course Learning Objectives

N/A

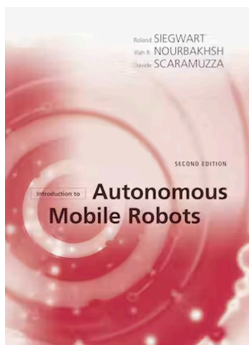
Assessment of Learning

Grading Components:

Grading Scale

Letter	Numerical Value
A	90 - 100
B	80 - 89
C	70 - 79
D	60 - 69
F	Below 60

Required Readings and Materials

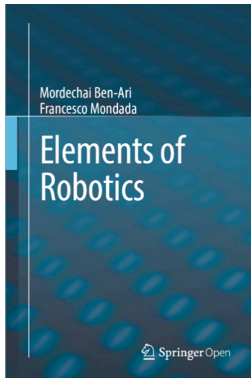


Introduction to Autonomous Mobile Robots

Authors: Roland Siegwart, Illah Reza Nourbakhsh, and Davide Scaramuzza

Edition: Second

Recommended Readings and Materials



Elements of Robotics

Authors: Mondada, F. and Mordechai B.

Edition: 2018



Handbook of Robotics

Authors: Bruno Siciliano, Oussama Khatib

Edition: 2008

Technology Needs

C++ programming

University Attendance Policy

[UTRGV's attendance policy](#)

Course Attendance

Attendance will be checked at random, and 5% of your credits will count toward your final grade. Students are expected to attend all scheduled classes and may be dropped from the course for excessive absences (please denote the specific number of unexcused absences that will trigger a “drop-by-instructor” in your class) UTRGV’s attendance policy excuses students from attending class if they are participating in officially sponsored university activities, such as athletics; have been provided such accommodation by Student Accessibility Services (SAS); for observance of religious holy days; or for military service.

University Policy on Recordings in Class

The use of classroom recordings is governed by the Federal Educational Rights and Privacy Act (FERPA), UTRGV's acceptable-use policy, and UTRGV HOP Policy STU 02-100 Student Conduct and Discipline. A recording of class sessions will be kept and stored by UTRGV, in accordance with FERPA and UTRGV policies. Your instructor will not share the recordings of your class activities outside of course participants, which include your fellow students, teaching assistants, or graduate assistants, and any guest faculty or community-based learning partners with whom we may engage during a class session. You may not share recordings outside of this course. As referenced in [UTRGV HOP Policy STU 02-100 Student Conduct and Discipline](#), doing so may result in disciplinary action. This is an instruction from university officials.

University AI Policy

At UTRGV, students are expected to use AI in ways that are honest, ethical, and consistent with course and assignment expectations. Because AI expectations may vary by course, instructor, assignment, and discipline, students should review the syllabus and assignment instructions before using AI. When expectations are unclear, students should ask their instructor before submitting work that involved AI. Students are responsible for checking the accuracy of AI-assisted work and protecting private or sensitive information.

Course Use of Artificial Intelligence (AI)

Here are the minimum requirements for using AI: 1. Please don't copy AI-generated content directly into your homework, reports, or projects without careful review and editing. 2. Please don't input non-public or confidential information into public AI tools.

University Academic Integrity Statement

Academic integrity is fundamental in our actions, as any act of dishonesty conflicts as much with academic achievement as with the values of honesty and integrity. Violations of academic integrity constitute academic dishonesty and include, but are not limited to: [cheating, plagiarism \(including self-plagiarism\), and collusion](#); submission for credit of any work or materials that are attributable in whole or in part to another person; taking an examination for another person; any act designed to give unfair advantage to a student; or the attempt to commit such acts ([UT System Board of Regents Rules and Regulations](#), [UTRGV Handbook of Operating Procedures STU 02-100](#), and [UTRGV Academic Integrity Guidelines](#)). **All violations of Academic Dishonesty will be reported to Student Rights and Responsibilities in accordance with [HOP STU 02-100, Student Code of Conduct](#).**

Course Academic Integrity Policy

Members of the UTRGV community uphold the Vaquero Honor Code's shared values of honesty, integrity, and mutual respect in our interactions and relationships. In this regard, academic integrity is fundamental in our actions, as any act of dishonesty conflicts as much with academic achievement as with the values of honesty and integrity. Violations of academic integrity include, but are not limited to: cheating, plagiarism (including self-plagiarism), and collusion; submission for credit of any work or materials that are attributable in whole or in part to another person; taking an examination for another person; any act designed to give unfair advantage to a student; or the attempt to commit such acts (Board of Regents Rules and Regulations, STU 02-100, and UTRGV Academic Integrity Guidelines). All violations of Academic Integrity will be reported to Student Rights and Responsibilities through Vaqueros Report It.

Student Support Resources

Student Support Services

Center Name	E-Mail
Advising Center	AcademicAdvising@utrgv.edu
Career Center	CareerCenter@utrgv.edu
Counseling Center	Counseling@utrgv.edu
Food Pantry	FoodPantry@utrgv.edu
Learning Center	LearningCenter@utrgv.edu
University Library	circulation@utrgv.edu
Writing Center	WC@utrgv.edu
UCentral	ucentral@utrgv.edu
Brightspace Assistance (COLTT)	Ticket Portal

University Policy Statements

SEXUAL MISCONDUCT AND MANDATORY REPORTING

In accordance with UT System and institutional policies, your instructor is a “Responsible Employee” for mandatory reporting purposes under Title IX regulations and Texas Education Code sections 51.251-252 and must report any instance of sexual misconduct by or against a student or employee of the institution, which includes sexual assault, stalking, dating violence, domestic violence, and sexual harassment to the [Office of Title IX and Equal Opportunity](#) (otixeo@utrgv.edu). More information can be found on the [OTIXEO website](#). If students, faculty, or staff would like confidential assistance, or have questions, they can contact OAVP ([Office for Advocacy & Violence Prevention](#)).

STUDENT ACCESSIBILITY SERVICES

[Student Accessibility Services](#) has offices on Brownsville and Edinburg campuses. [Visit the SAS web page to learn more and explore accessibility services.](#)

STUDENTS WITH DISABILITIES

Students with a documented disability (physical, psychological, learning, or other disability which affects academic performance) who would like to receive reasonable academic accommodations should contact [Student Accessibility Services \(SAS\)](#) for additional information. The

student must apply for accommodations using the [mySAS portal](#) and is responsible for providing sufficient documentation of the disability to SAS. Upon submission of the request, students should expect to participate in an interactive discussion, or an intake appointment, with SAS staff. Accommodations may be requested at any time but are not granted retroactively, meaning they are valid moving forward after approval by SAS. Academic accommodations will not apply prior to SAS' approval. Students should contact SAS early in the semester/module for guidance.

Students who experience a broken bone, severe injury, or undergo surgery may also be eligible for temporary accommodations. Please contact [Student Accessibility Services \(SAS\)](#) for more information.

PREGNANCY, PREGNANCY-RELATED, AND PARENTING ACCOMMODATIONS

Title IX of the Education Amendments of 1972 prohibits discrimination based on sex, which includes discrimination based on pregnancy, marital status, or parental status.

Students seeking accommodations related to pregnancy, pregnancy-related condition, or parenting should submit the request using the form found at [Pregnancy and Parenting | UTRGV](#).

MANDATORY COURSE EVALUATION PERIOD

Students have the opportunity to complete an ONLINE evaluation of this course through Watermark Course Evaluations and Surveys, which may be accessed through my.UTRGV or the Brightspace course module. Course evaluations are used by the instructor to inform revisions of the course to ensure student success. Course evaluations are also used by the instructor for annual performance review, promotion applications, teaching award applications, among others.